Computer Organization Homework 3 (Due on 2023/11/14)

1. What decimal number does the bit pattern 0x0c000000 represent if it is a floating point number? Use the IEEE 754 single precision. (10%)

2. What is the largest positive number and smallest positive non-zero number in IEEE754 single precision? What is the smallest number in IEEE754 single precision ? The answer cannot be negative infinite(10%)

3. Please briefly explain Guard bit ,Round bit and Sticky bit then show how IEEE754 single precision 44454400 + 451FE980 round to first decimal places with these three bits step by step.(30%)

4. Assume 185 and 122 are unsigned 8-bit decimal integers. Calculate 185 - 122. Is there overflow, underflow, or neither? Why?(10%)

5. Fill the blank (mark by yellow) in the table if it contains the machine codes for the following MIPS assembly instructions. (You can express in decimal values.) (30%)

```
LOOP1:  sll   $s3, $t1, 3
        slt   $s2, $s3, $t2
        beq   $s2, $zero, EXIT
        sub   $s6, $s6, $t1
        sw    $s5, 24($s3)
        addi  $t1, $t1, 8
        j     LOOP1
EXIT:
```

Address	OP					
28244	0	0	(1)	(2)	(3)	0 (sll)
28248	0	(4)	10	(5)	0	42 (slt)
28252	4 (beq)	18	0	(6)		
28256	0	22	9	22	0	34 (sub)
28260	43(sw)	19	21	(7)		
28264	8 (addi)	(8)	(8)	(9)		
28268	2 (j)	(10)				
28272	(EXIT)					

6. Write down the binary representation of the decimal number 119.25 assuming the IEEE 754 single precision format.(10%)